Predicting Weekly Work Hours
Micro-Project 3

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Problem Statement

- Adults work different numbers of hours each week.
- Question: How accurately can age, education, occupation, and workclass predict weekly work hours?
- Module 2 examined income categories. This module changes the outcome to hours because the assignment calls for regression and the file has no actual salary amounts.
- The goal is to compare predictions on records the models did not use for training.

Hypothesis Formulation

- Age, education, occupation, and workclass together will predict weekly hours more accurately than the training-sample average alone.
- Possible explanation: Job roles and employment arrangements differ, which may help explain differences in work hours.
- Select settings and a preferred model using training cross-validation RMSE. Then evaluate the fixed models on a separate test set.
- A predictive improvement would support this hypothesis within the sample. It would not establish a causal explanation.

Acquire

- Source: UCI Adult dataset, drawn from the 1994 Census database.
- 32,561 records and 15 fields from the supplied training sample. The separate UCI test file is not used.
- The unchanged CSV is included with the Jupyter notebook in the Week 3 folder.
- Tools used: Jupyter Notebook, Python, Pandas, NumPy, Matplotlib, and scikit-learn.
- Becker & Kohavi (1996), Adult. https://doi.org/10.24432/C5XW20

Prepare: Data and Predictors

- Trim spaces, mark missing values, and check numeric ranges, income labels, and education codes. Keep all 32,561 records.
- Keep original age and hours. Use education, occupation, and workclass as categories. Education codes are not years of schooling.
- Missing workclass (1,836) and occupation (1,843) become Unknown in the model inputs.
- Exclude income, fnlwgt, duplicate education codes, and every hour-derived field from the predictors. Keep fnlwgt in the source data.
- Each row counts equally. Results do not estimate population work hours.

Prepare: Training and Test Records

- 26,048 training records and 6,513 test records, approximately an 80/20 split. Random seed: 42.
- Keep identical full profiles together, including 24 repeated profiles beyond the first occurrence. They never cross split or validation boundaries.
- Use the same three training-validation folds for every model. Learn scaling and category encoding inside each fold.
- Choose model settings and the preferred model before looking at test performance. Keep the test results out of tuning.

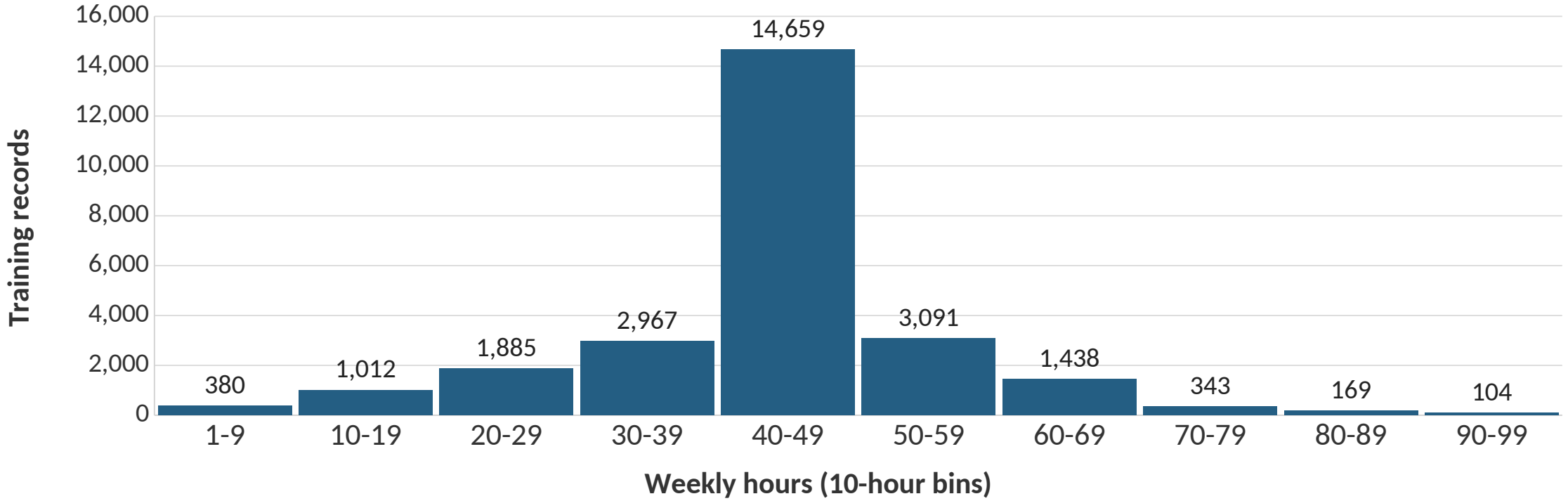
Analyze: Regression Methods

Model	Approach	Settings compared
Mean baseline	Predict training mean	No tuning
Linear regression	Linear relationships	No tuning
SVR	RBF kernel; nonlinear	C: 1, 10; epsilon: 0.5, 2
Decision tree	Branching rules	Depth: 3, 5, 8; leaf: 25, 100

The mean baseline uses no predictor information. Every model is evaluated on the same records.

Source: UCI Adult, 1994 sample. Results are unweighted.

Analyze: Training Work Hours



46.8% of training records report exactly 40 hours. Mean: 40.39 hours; median: 40.

Source: UCI Adult, 1994 sample. Results are unweighted.

Analyze: Cross-Validation Results

Model	CV RMSE (hours)	Fold SD (hours)
Mean baseline	12.340	0.240
Linear regression	11.542	0.199
SVR	10.926	0.187
Decision tree	11.021	0.167

Three folds from training data only. Fold SD describes variation; it is not a confidence interval.

Support vector regression had the lowest mean validation RMSE. Selected SVR settings: C = 10, epsilon = 2 hours, RBF kernel.

Source: UCI Adult, 1994 sample. Results are unweighted.

Report: Final Test Results

Model	MAE (hours)	RMSE (hours)	R-squared
Mean baseline	7.61	12.37	-0.0003
Linear regression	7.87	11.57	0.1257
SVR	7.16	10.89	0.2250
Decision tree	7.49	10.96	0.2142

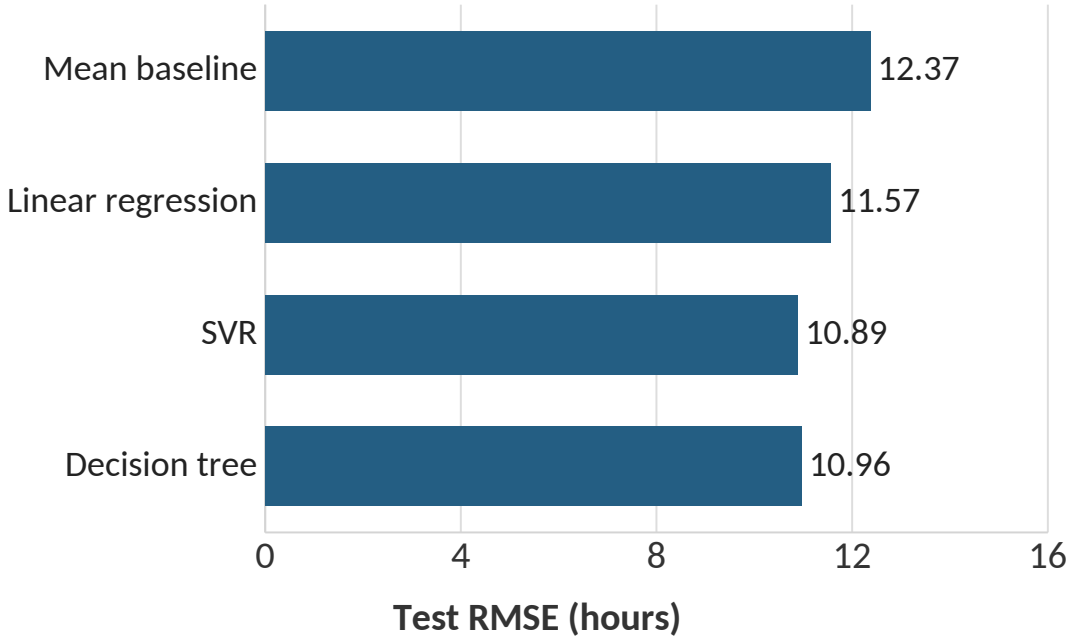
Same 6,513 test records. Lower MAE/RMSE is better; higher R-squared is better.

SVR test RMSE is 12.0% below the mean baseline. Model selection used cross-validation.

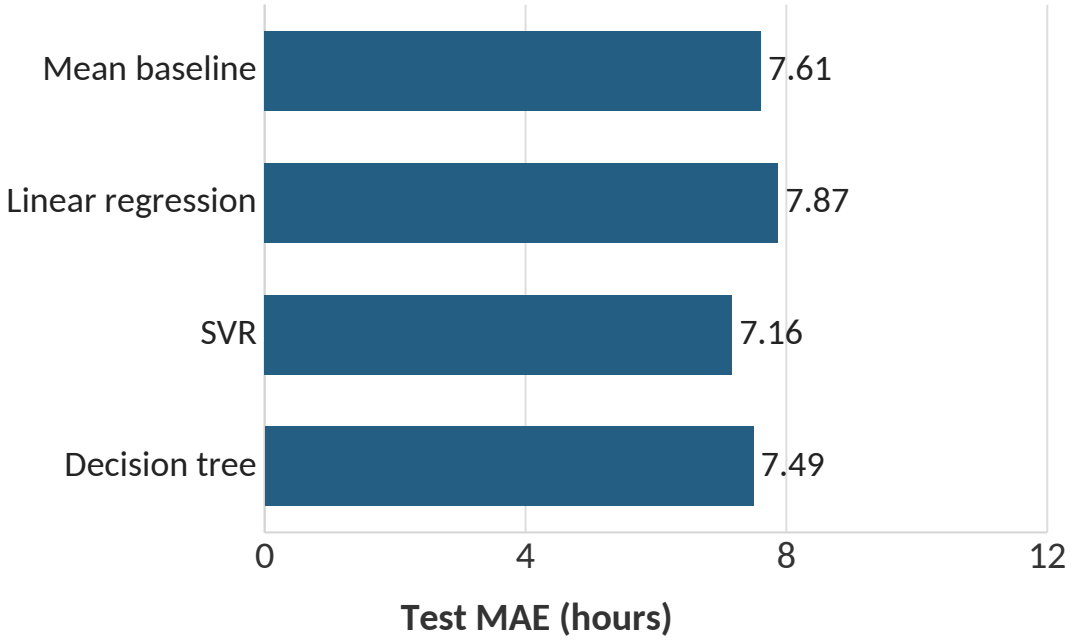
Source: UCI Adult, 1994 sample. Results are unweighted.

Report: Prediction Errors

Root mean squared error



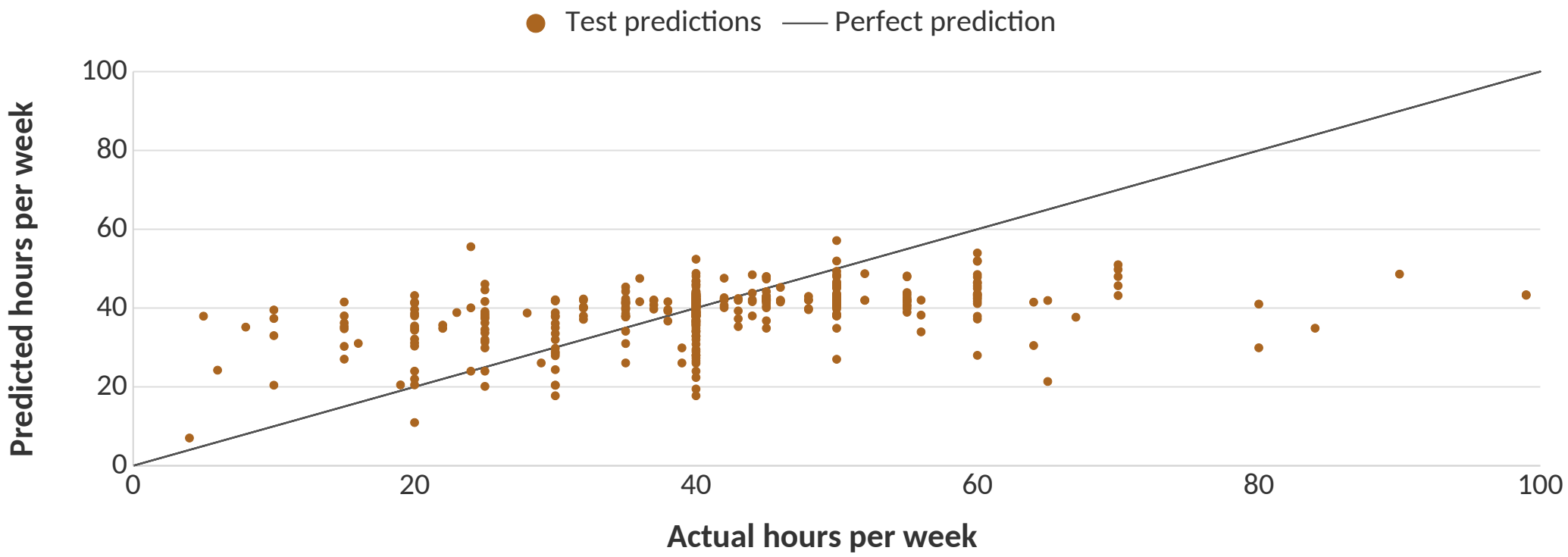
Mean absolute error



MAE describes the average absolute miss. RMSE gives more weight to large errors. Both are measured in hours.

Source: UCI Adult, 1994 sample. Results are unweighted.

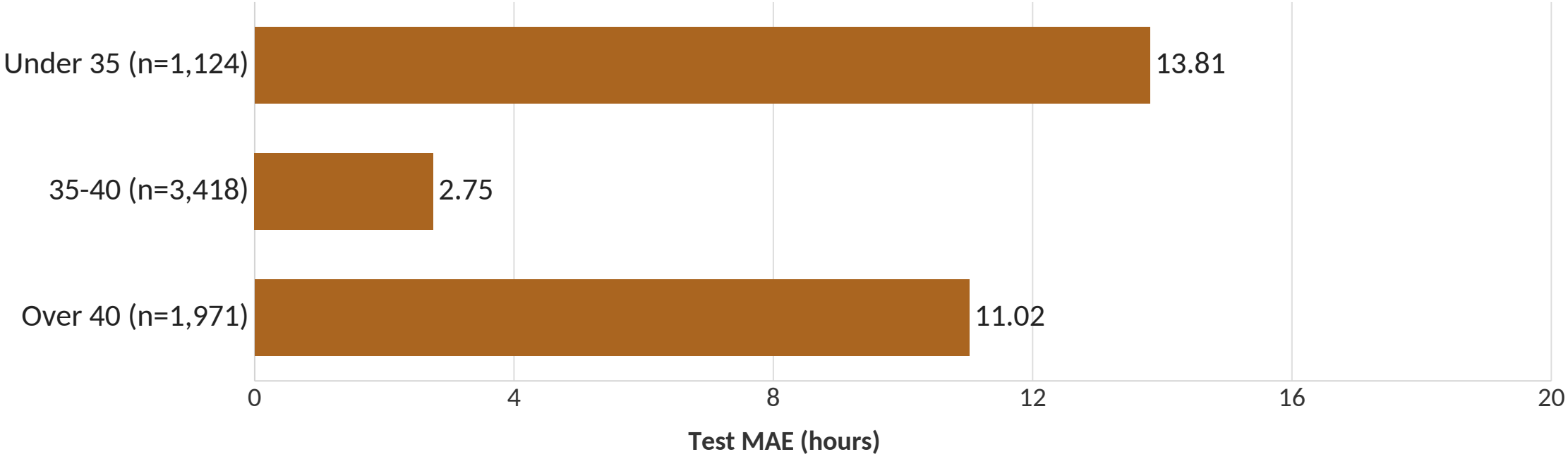
Report: Predicted and Actual Hours



Selected model: Support vector regression. Points show a fixed sample of 500 test records; scores use all 6,513.

Source: UCI Adult, 1994 sample. Results are unweighted.

Report: Error by Actual Workweek



The largest average absolute error is in the under 35 group (13.81 hours). Bands are used only to inspect errors.

Source: UCI Adult, 1994 sample. Results are unweighted.

Report: Findings and Limitations

- The selected support vector regression improves test RMSE from 12.37 to 10.89 hours. This supports the predictive hypothesis within this sample.
- Its average absolute miss is 7.16 hours, and R-squared is 0.225. Much of the variation remains unexplained.
- Historical, unweighted records do not establish current population patterns or causal effects. Missing categories and omitted factors remain limitations.
- The search was limited, and this is one internal test split from data explored in earlier modules. Small score differences are not proof of superiority.

Act: Apply the Selected Model

Age	Education	Occupation	Workclass	Actual hours	Predicted hours
49	9th	Other-service	Private	16	38.30
31	Masters	Prof-specialty	Private	50	42.27
23	Assoc-acdm	Protective-serv	Local-gov	52	40.43

Examples are the first three held-out records by source order. Predictions are estimates, not individual scheduling recommendations.

Source: UCI Adult, 1994 sample. Results are unweighted.

Act: Decision and Next Steps

- Keep support vector regression as the preferred model in this comparison because it had the lowest training-validation RMSE.
- Applying it to held-out records shows improvement over the baseline, but an average miss of 7.16 hours is too large to treat as a precise personal schedule.
- Use these results to guide further study of work-hour differences. Do not use this historical classroom model for individual employment decisions.
- Before practical use, define an acceptable error, obtain current independent data, and assess errors across relevant groups. Study survey weights if making population claims.
- This micro-project completes Step 5, Act.